



7.0.1 ISRO CSE 2009 | Question: 69

The 'command' used to change contents of one database using the contents of another database by linking them on a common key field?

- A. Replace B. Join C. Change D. Update

isro2009 databases bad-question

Answer key

7.0.2 ISRO-DEC2017-14

Type-IV *JDBC* driver is a driver

- A. which is written in *C++*
 B. which requires an intermediate layer
 C. which communicates through Java sockets
 D. which translates *JDBC* function calls into *API* not native to *DBMS*

isrodec2017 databases

Answer key

7.1 B Tree (6)

7.1.1 B Tree: GATE CSE 2007 | Question: 63, ISRO2016-59

The order of a leaf node in a B^+ - tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is $1K$ bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

- A. 63 B. 64 C. 67 D. 68

gatecse-2007 databases b-tree normal isro2016

Answer key

7.1.2 B Tree: GATE IT 2005 | Question: 23, ISRO2017-67

A B-Tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are

- A. 5 B. 4 C. 3 D. 2

gateit-2005 databases b-tree normal isro2017

Answer key

7.1.3 B Tree: ISRO CSE 2018 | Question: 49

in a file which contains 1 million records and the order of the tree is 100 , then what is the maximum number of nodes to be accessed if B^+ tree index is used?

- a. 5 b. 4 c. 3 d. 10

isro2018 databases b-tree

Answer key

7.1.4 B Tree: ISRO-2013-26



Calculate the order of leaf (P_{leaf}) and non leaf (P) nodes of a B^+ tree based on the information given below.

Search key field = 12 field

Record pointer = 10 bytes

Block pointer = 8 bytes

Block size = 1KB

A. $P_{leaf} = 51$ & $p = 46$

C. $P_{leaf} = 46$ & $p = 51$

B. $P_{leaf} = 47$ & $p = 52$

D. $P_{leaf} = 52$ & $p = 47$

isro2013 databases b-tree

Answer key

7.1.5 B Tree: ISRO-2013-27



The physical location of a record determined by a formula that transforms a file key into a record location is

A. Hashed file

B. B -Tree file

C. Indexed file

D. Sequential file

isro2013 databases b-tree

Answer key

7.1.6 B Tree: ISRO2015-39



If a node has K children in B tree, then the node contains exactly _____ keys.

A. K^2

B. $K - 1$

C. $K + 1$

D. $\sqrt{K}$

isro2015 databases b-tree

Answer key

7.2 Bit Mapped Index (1)

7.2.1 Bit Mapped Index: ISRO-DEC2017-12



Consider a table that describes the customers:

Customers(custid, name, gender, rating)

The rating value is an integer in the range 1 to 5 and only two values (male and female) are recorded for *gender*. Consider the query "how many male customers have a rating of 5" ? The best indexing mechanism appropriate for the query is

A. Linear hashing

B. Extendible hashing

C. B+ tree

D. Bit-mapped index

isrodec2017 databases indexing bit-mapped-index

Answer key

7.3 Candidate Key (1)

7.3.1 Candidate Key: ISRO CSE 2014 | Question: 42



Let x, y, z, a, b, c be the attributes of an entity set E . If $\{x\}, \{x, y\}, \{a, b\}, \{a, b, c\}, \{x, y, z\}$ are superkeys then which of the following are the candidate keys?

- A. $\{x, y\}$ and $\{a, b\}$
- B. $\{x\}$ and $\{a, b\}$
- C. $\{x, y, z\}$ and $\{a, b, c\}$
- D. $\{z\}$ and $\{c\}$

databases candidate-key isro2014

Answer key 

7.4 Database Normalization (13)

7.4.1 Database Normalization: ISRO CSE 2007 | Question: 51



Armstrong's inference rule doesnot determine

- A. Reflexivity
- B. Augmentation
- C. Transitivity
- D. Mutual dependency

isro2007 databases database-normalization

Answer key 

7.4.2 Database Normalization: ISRO CSE 2007 | Question: 54



BCNF is not used for cases where a relation has

- A. Two (or more) candidate keys
- B. Two candidate keys and composite
- C. The candidate key overlap
- D. Two mutually exclusive foreign keys

isro2007 databases database-normalization

Answer key 

7.4.3 Database Normalization: ISRO CSE 2011 | Question: 29



Which normal form is based on the concept of 'full functional dependency' is

- A. First Normal Form
- B. Second Normal Form
- C. Third Normal Form
- D. Fourth Normal Form

isro2011 databases database-normalization

Answer key 

7.4.4 Database Normalization: ISRO CSE 2011 | Question: 53



In functional dependency Armstrong inference rules refers to

- A. Reflexive, Augmentation and Decomposition
- B. Transitive, Augmentation and Reflexive
- C. Augmentation, Transitive, Reflexive and Decomposition
- D. Reflexive, Transitive and Decomposition

isro2011 databases database-normalization

Answer key 

7.4.5 Database Normalization: ISRO CSE 2014 | Question: 22



Consider the schema $R(A, B, C, D)$ and the functional dependencies $A \rightarrow B$ and $C \rightarrow D$. If the decomposition is made as $R_1(A, B)$ and $R_2(C, D)$, then which of the following is TRUE?

- A. Preserves dependency but cannot perform lossless join
- B. Preserves dependency and performs lossless join
- C. Does not perform dependency and cannot perform lossless join

D. Does not preserve dependency but perform lossless join

databases isro2014 database-normalization

Answer key

7.4.6 Database Normalization: ISRO CSE 2014 | Question: 30



Every time the attribute A appears, it is matched with the same value of attribute B but not the same value of attribute C. Which of the following is true?

- A. $A \rightarrow (B, C)$
- B. $A \rightarrow B, A \twoheadrightarrow C$
- C. $A \rightarrow B, C \twoheadrightarrow A$
- D. $A \twoheadrightarrow B, B \rightarrow C$

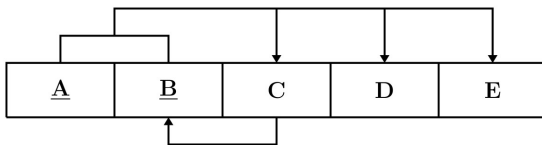
databases database-normalization isro2014

Answer key

7.4.7 Database Normalization: ISRO CSE 2014 | Question: 63



Consider the following table



The table is in which normal form?

- A. First Normal Form
- B. Second Normal Form
- C. Third Normal Form but not BCNF
- D. Third Normal Form and BCNF

isro2014 databases database-normalization

Answer key

7.4.8 Database Normalization: ISRO CSE 2018 | Question: 41



For a database relation $R(a, b, c, d)$ where the domain of a, b, c and d include only atomic value, only the following functions dependencies and those that can be inferred from them hold

$$a \rightarrow c$$

$$b \rightarrow d$$

The relation is in:

- a. First normal form but not in second normal form
- b. Second normal form but not in third normal form
- c. Third normal form
- d. None of the above

isro2018 database-normalization databases

Answer key

7.4.9 Database Normalization: ISRO CSE 2018 | Question: 5



Considering the following table in a relational database :

Last Name	Rank	Room	shift
Smith	Manager	234	Morning
Jones	Custodian	33	Afternoon
Smith	Custodian	33	Evening
Doe	Clerical	222	Morning

According to the data shown in the table, which of the following could be a candidate key of the table?

- a. {Last Name} b. {Room} c. {Shift} d. {Room, Shift}

isro2018 databases database-normalization

Answer key 

7.4.10 Database Normalization: ISRO CSE 2018 | Question: 54



The set of attributes X will be fully functionally dependent on the set of attributes Y if the following conditions are satisfied.

- a. X is functionally dependent on Y b. X is not functionally dependent on any subset of Y
c. Both (a) and (b) d. None of these

isro2018 databases database-normalization

Answer key 

7.4.11 Database Normalization: ISRO CSE 2020 | Question: 50



If every non-key attribute functionally dependent on the primary key, then the relation will be in

- A. First normal form B. Second normal form
C. Third normal form D. Fourth Normal form

isro-2020 databases database-normalization easy

Answer key 

7.4.12 Database Normalization: ISRO CSE 2023 | Question: 50



Fifth Normal form is concerned with

- A. Join dependency B. Domain-key
C. Multivalued dependency D. Functional dependency

isro-cse-2023 database-normalization databases

Answer key 

7.4.13 Database Normalization: ISRO-2013-25



Consider the following dependencies and the BOOK table in a relational database design. Determine the normal form of the given relation.

ISBN $\rightarrow$ Title

ISBN $\rightarrow$ Publisher

Publisher $\rightarrow$ Address

- A. First Normal Form B. Second Normal Form
C. Third Normal Form D. BCNF

isro2013 databases database-normalization

Answer key 

7.5 Distributed Computing (1)

7.5.1 Distributed Computing: ISRO CSE 2023 | Question: 35



Which type of data storage system Cassandra is?

- A. Distributed B. Centralized C. Parallel D. Dumb

Answer key

7.6

Division (1)

7.6.1 Division: ISRO-DEC2017-13



Consider the following schema:

Sailors(sid,sname,rating,age)

Boats(bid,bname,colour)

Reserves(sid,bid,day)

Two boats can have the same name but the colour differentiates them.

The two relations

$$\rho(Tempside, (\pi_{sid,bid} Reserves) / (\pi_{bid} (\sigma_{bname \neq Ganga} Boats))),$$

$$\pi_{sname} (Tempside \bowtie Sailors)$$

If / is division operation, the above set of relations represents the query

- A. Names of sailors who have reserved all boats called Ganga.
- B. Names of sailors who have not reserved any Ganga boat.
- C. Names of sailors who have reserved at least one Ganga boat.
- D. Names of sailors who have reserved at most one Ganga boat.

isrodec2017 relational-algebra databases division joins

Answer key

7.7

ER Diagram (5)

7.7.1 ER Diagram: ISRO CSE 2016 | Question: 56



Goals for the design of the logical scheme include

- A. avoiding data inconsistency
- B. being able to construct query easily
- C. being able to access data efficiently
- D. All of the above

isro2016 databases er-diagram

Answer key

7.7.2 ER Diagram: ISRO CSE 2017 | Question: 8



Which symbol denote derived attributes in ER Model?

- A. Double ellipse
- B. Dashed ellipse
- C. Squared ellipse
- D. Ellipse with attribute name underlined

isro2017 databases er-diagram

Answer key

7.7.3 ER Diagram: ISRO CSE 2018 | Question: 52



In E-R model, Y is the dominant entity and X is subordinate entity

- a. If X is deleted, then Y is also deleted
- b. If Y is deleted, then X is also deleted
- c. If Y is deleted, then X is not deleted

d. None of the above

isro2018 er-diagram databases

Answer key

7.7.4 ER Diagram: ISRO CSE 2023 | Question: 62



Entity Relationship model shows the

- A. Static view
- B. Functional view
- C. Dynamic view
- D. All of the above

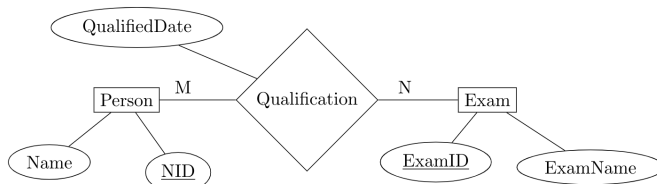
isro-cse-2023 databases er-diagram

Answer key

7.7.5 ER Diagram: ISRO2015-16



Consider the following Relationship Entity Diagram(ERP)



Which of the following possible relations will not hold if the above ERD is mapped into a relation model?

- A. Person (NID, Name)
- B. Qualification (NID, ExamID, QualifiedDate)
- C. Exam (ExamID, NID, ExamName)
- D. Exam (ExamID, ExamName)

isro2015 databases er-diagram

Answer key

7.8 File System (6)

7.8.1 File System: ISRO CSE 2009 | Question: 70



A locked database file can be

- A. Accessed by only one user
- B. Modified by users with the correct password
- C. Used to hide sensitive information
- D. Updated by more than one user

isro2009 databases file-system

Answer key

7.8.2 File System: ISRO CSE 2011 | Question: 46



Which RAID level gives block level striping with double distributed parity?

- A. RAID 10
- B. RAID 2
- C. RAID 6
- D. RAID 5

isro2011 file-system

Answer key

7.8.3 File System: ISRO CSE 2017 | Question: 4



What does a data dictionary will identify?

- A. Field name B. Field format C. Field types D. All of the above

isro2017 databases file-system

Answer key 

7.8.4 File System: ISRO CSE 2017 | Question: 45



Which of these is characteristic of RAID 5?

- A. Dedicated parity B. Double parity
C. Hamming code parity D. Distributed parity

isro2017 databases file-system non-gatecse

Answer key 

7.8.5 File System: ISRO CSE 2018 | Question: 53



Immunity of the external schemas (or application programs) to changes in the conceptual scheme is referred to as:

- a. Physical Data Independence b. Logical Data Independence
c. Both (a) and (b) d. None of the above

isro2018 databases file-system

Answer key 

7.8.6 File System: ISRO2015-13



Six files F1, F2, F3, F4, F5 and F6 have 100, 200, 50, 80, 120, 150 records respectively. In what order should they be stored so as to optimize act. Assume each file is accessed with the same frequency

- A. F3, F4, F1, F5, F6, F2
B. F2, F6, F5, F1, F4, F3
C. F1, F2, F3, F4, F5, F6
D. Ordering is immaterial as all files are accessed with the same frequency.

isro2015 file-system

Answer key 

7.9 Functional Dependency (1)

7.9.1 Functional Dependency: ISRO2015-20



Let $R = (A, B, C, D, E, F)$ be a relation scheme with the following dependencies $C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$. Which of the following is a key of R?

- A. CD B. EC C. AE D. AC

isro2015 databases database-normalization functional-dependency

Answer key 

7.10 Indexing (4)

7.10.1 Indexing: GATE CSE 2008 | Question: 16, ISRO2016-60



A clustering index is defined on the fields which are of type

- A. non-key and ordering B. non-key and non-ordering

C. key and ordering

D. key and non-ordering

gatecse-2008 easy databases indexing isro2016

Answer key

7.10.2 Indexing: ISRO CSE 2018 | Question: 51



Which of the following is dense index?

- A. Primary index
- C. Secondary index

- B. Clustered index
- D. Secondary non-key index

isro2018 indexing

Answer key

7.10.3 Indexing: ISRO-2013-61, ISRO CSE 2008 | Question: 35



Embedded pointer provides

- A. A secondary access path
- C. An inverted index

- B. A physical record key
- D. A primary key

isro2013 databases indexing isro2008

Answer key

7.10.4 Indexing: ISRO2015-18



Given a block can hold either 3 records or 10 key pointers. A database contains n records, then how many blocks do we need to hold the data file and the dense index

- A. $13n/30$
- B. $n/3$
- C. $n/10$
- D. $n/30$

isro2015 databases indexing

Answer key

7.11 Joins (1)

7.11.1 Joins: ISRO CSE 2023 | Question: 27



In SQL the statement $\text{select } * \text{ from } R, S$ is equivalent to

- A. $\text{Select } * \text{ from } R \text{ natural join } S$
- B. $\text{Select } * \text{ from } R \text{ cross join } S$
- C. $\text{Select } * \text{ from } R \text{ outer join } S$
- D. $\text{Select } * \text{ from } R \text{ inner join } S$

isro-cse-2023 databases joins easy

Answer key

7.12 Normal Forms (1)

7.12.1 Normal Forms: ISRO-DEC2017-15



Consider the following table: Faculty(facName, dept, office, rank, dateHired)

facName	dept	office	rank	dateHired
Ravi	Art	A101	Professor	1975
Murali	Math	M201	Assistant	2000

facName	dept	office	rank	dateHired
Narayanan	Art	A101	Associate	1992
Lakshmi	Math	M201	Professor	1982
Mohan	CSC	C101	Professor	1980
Sreeni	Math	M203	Associate	1990
Tanuja	CSC	C101	Instructor	2001
Ganesh	CSC	C105	Associate	1995

(Assume that no faculty member within a single department has same name. Each faculty member has only one office identified in *office*). *3NF* refers to third normal form and *BCNF* refers to Boyce-Codd Normal Form

Then *Faculty* is

- A. Not in 3NF, in BCNF
- B. In 3NF, not in BCNF
- C. In 3NF, in BCNF
- D. Not in 3NF, not in BCNF

isrodec2017 databases database-normalization normal-forms

Answer key 

7.13

Query (2)

7.13.1 Query: ISRO CSE 2023 | Question: 28



Which of the following SQL query deletes all tuples in the teacher relation for those teachers associated with a department located in the building with name CSE and which is in department relation:

- a. DELETE FROM teacher
WHERE dept_name IN 'CSE';
- b. DELETE FROM department
WHERE building = 'CSE';
- c. DELETE FROM teacher
WHERE dept_name IN (
SELECT dept_name
FROM department
WHERE building = 'CSE');
- d. DELETE FROM teacher
WHERE dept_name IN 'CSE';

isro-cse-2023 databases sql query

Answer key 

7.13.2 Query: ISRO-DEC2017-16



Consider the following query :

```
SELECT E.eno, COUNT(*)
FROM Employees E
GROUP BY E.eno
```

If an index on *eno* is available, the query can be answered by scanning only the index if

- A. the index is only hash and clustered
- B. the index is only *B+* tree and clustered
- C. index can be hash or *B+* tree and clustered or non-clustered

D. index can be hash or $B+$ tree and clustered

isrodec2017 sql databases indexing query

Answer key

7.14

Rdbms (2)

7.14.1 Rdbms: ISRO CSE 2007 | Question: 50



A view of database that appears to an application program is known as

- A. Schema B. Subschema C. Virtual table D. None of these

isro2007 databases rdbms

Answer key

7.14.2 Rdbms: ISRO CSE 2011 | Question: 21



Which type of DBMS provides support for maintaining several versions of the same entity?

- A. Relational Database Management System B. Hierarchical
C. Object Oriented Database Management System D. Network

isro2011 databases rdbms

Answer key

7.15

Referential Integrity (2)

7.15.1 Referential Integrity: GATE CSE 1997 | Question: 6.10, ISRO2016-54



Let $R(a, b, c)$ and $S(d, e, f)$ be two relations in which d is the foreign key of S that refers to the primary key of R . Consider the following four operations R and S

- I. Insert into R
II. Insert into S
III. Delete from R
IV. Delete from S

Which of the following can cause violation of the referential integrity constraint above?

- A. Both I and IV B. Both II and III C. All of these D. None of these

gate1997 databases referential-integrity easy isro2016

Answer key

7.15.2 Referential Integrity: ISRO CSE 2009 | Question: 72



Purpose of 'Foreign Key' in a table is to ensure

- A. Null Integrity B. Referential Integrity
C. Domain Integrity D. Null and Domain Integrity

isro2009 databases referential-integrity

Answer key

7.16

Relational Algebra (4)

7.16.1 Relational Algebra: GATE CSE 1999 | Question: 1.18, ISRO2016-53



Consider the join of a relation R with a relation S . If R has m tuples and S has n tuples then the maximum and minimum sizes of the join respectively are

- A. $m + n$ and 0
- B. mn and 0
- C. $m + n$ and $|m - n|$
- D. mn and $m + n$

gate1999 databases relational-algebra easy isro2016

Answer key

7.16.2 Relational Algebra: GATE CSE 2000 | Question: 1.23, ISRO2016-57



Given the relations

- employee (name, salary, dept-no), and
- department (dept-no, dept-name, address),

Which of the following queries cannot be expressed using the basic relational algebra operations ($\sigma, \pi, \times, \bowtie, \cup, \cap, -$)?

- A. Department address of every employee
- B. Employees whose name is the same as their department name
- C. The sum of all employees' salaries
- D. All employees of a given department

gatecse-2000 databases relational-algebra easy isro2016

Answer key

7.16.3 Relational Algebra: ISRO CSE 2007 | Question: 52



Which operation is used to extract specified columns from a table?

- A. Project
- B. Join
- C. Extract
- D. Substitute

isro2007 databases relational-algebra

Answer key

7.16.4 Relational Algebra: ISRO CSE 2008 | Question: 33



The join operation can be defined as

- A. a cartesian product of two relations followed by a selection
- B. a cartesian product of two relations
- C. a union of two relations followed by cartesian product of the two relations
- D. a union of two relations

isro2008 databases relational-algebra

Answer key

7.17.1 Relational Model: ISRO CSE 2014 | Question: 11



An aggregation association is drawn using which symbol?

- A. A line which loops back on to the same table
- B. A small open diamond at the end of a line connecting two tables
- C. A small closed diamond at the end of a line connecting two tables
- D. A small closed triangle at the end of a line connecting two tables

isro2014 databases relational-model

Answer key

7.17.2 Relational Model: ISRO2015-21



If $D_1, D_2, \dots, D_n$ are domains in a relational model, then the relation is a table, which is a subset of

- A. $D_1 \oplus D_2 \oplus \dots \oplus D_n$
- B. $D_1 \times D_2 \times \dots \times D_n$
- C. $D_1 \cup D_2 \cup \dots \cup D_n$
- D. $D_1 \cap D_2 \cap \dots \cap D_n$

databases isro2015 relational-model

Answer key

7.18

SQL (14)

7.18.1 SQL: GATE CSE 2005 | Question: 77, ISRO2016-55



The relation **book** (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

```
select title
from book as B
where (select count(*)
      from book as T
      where T.price > B.price) < 5
```

- A. Titles of the four most expensive books
- B. Title of the fifth most inexpensive book
- C. Title of the fifth most expensive book
- D. Titles of the five most expensive books

gatecse-2005 databases sql easy isro2016

Answer key

7.18.2 SQL: ISRO CSE 2007 | Question: 47



Which commands are used to control access over objects in relational database?

- A. CASCADE & MVD
- B. GRANT & REVOKE
- C. QUE & QUIST
- D. None of these

isro2007 databases sql

Answer key

7.18.3 SQL: ISRO CSE 2007 | Question: 48



Which of the following is aggregate function in SQL?

- A. Avg
- B. Select
- C. Ordered by
- D. distinct

Answer key

7.18.4 SQL: ISRO CSE 2017 | Question: 7



Employee	Department	OT_allowance
RAMA	Mechanical	5000
GOPI	Electrical	2000
SINDHU	Computer	4000
MAHESH	Civil	1500

What is the output of the following SQL query?

```
select count(*) from
((select Employee, Department from Overtime_allowance)
natural join
(select Department, OT_allowance from Overtime_allowance)
as T);
```

- A. 16 B. 4 C. 8 D. None of the above

Answer key

7.18.5 SQL: ISRO CSE 2018 | Question: 40



Given relations $R(w,x)$ and $S(y,z)$, the result of

```
SELECT DISTINCT w, x
FROM R, S
```

Is guaranteed to be same as R , if

- a. R has no duplicates and S is non-empty b. R and S have no duplicates
c. S has no duplicates and R is non-empty d. R and S have the same number of tuples

Answer key

7.18.6 SQL: ISRO CSE 2018 | Question: 42



Consider the set of relations given below and the SQL query that follows:

```
Students: (Roll_number, Name, Date_of_birth)
Coursed: (Course_number, Course_name, Instructor)
Grades: (Roll_number, Course_number, Grade)
```

```
SELECT DISTINCT Name
FROM Students, Courses, Grades
WHERE Students.Roll_number = Grades.Roll_number
      AND Courses.Instructor = Sriram
      AND Courses.Course_number = Grades.Course_number
      AND Grades.Grade = A
```

Which of the following sets is computed by the above query?

- a. Names of Students who have got an A grade in all courses taught by Sriram
b. Names of Students who have got an A grade in all courses
c. Names of Students who have got an A grade in at least one of the courses taught by Sriram
d. None of the above

Answer key

7.18.7 SQL: ISRO CSE 2020 | Question: 48



Properties of 'DELETE' and 'TRUNCATE' commands indicate that

- A. After the execution of 'TRUNCATE' operation, COMMIT, and ROLLBACK statements cannot be performed to retrieve the lost data, while 'DELETE' allow it
- B. After the execution of 'DELETE' and 'TRUNCATE' operation retrieval is easily possible for the lost data
- C. After the execution of 'DELETE' operation, COMMIT and ROLLBACK statements can be performed to retrieve the lost data, while TRUNCATE do not allow it
- D. After the execution of 'DELETE' and 'TRUNCATE' operation no retrieval is possible for the lost data

Answer key

7.18.8 SQL: ISRO CSE 2020 | Question: 51



The SQL query

```
SELECT columns
FROM TableA
RIGHT OUTER JOIN TableB
ON A.columnName = B.columnName
WHERE A.columnName IS NULL
```

returns the following:

- A. All rows in Table B, which meets equality condition above and, none from Table A which meets the condition.
- B. All rows in Table A, which meets equality condition above and none from Table B, which meets the condition.
- C. All rows in Table B, which meets the equality condition
- D. All rows in Table A, which meets the equality condition

Answer key

7.18.9 SQL: ISRO CSE 2023 | Question: 7



Consider a database table T containing two columns X and Y each of type Integer. After the creation of the table, one record ($X = 1, Y = 1$) is inserted in the table. Let MX and MY denote the respective maximum values of X and Y among all records in the table at any point in time. Using MX and MY , new records are inserted in the table 128 times with X and Y values being $MX + 1, 2 \times MY + 1$ respectively. It may be noted that each time after the insertion, values of MX and MY change. What will be the output of the following SQL query after the steps mentioned above are carried out?

SELECT Y FROM T WHERE $X = 7$;

- A. 127
- B. 15
- C. 129
- D. 257

Answer key



7.18.10 SQL: ISRO-2013-24

Consider the following relational schema:

- Suppliers (sid:integer, sname:string, saddress:string)
- Parts (pid:integer, pname:string, pcolor:string)
- Catalog (sid:integer, pid:integer, pcost:real)

What is the result of the following query?

```
(SELECT Catalog.pid from Suppliers, Catalog
WHERE Suppliers.sid = Catalog.sid)
MINUS
(SELECT Catalog.pid from Suppliers, Catalog
WHERE Suppliers.sname <> 'Sachin' and Suppliers.sid = Catalog.sid)
```

- A. Pid of parts supplied by all except Sachin
- B. Pid of parts supplied only by Sachin
- C. Pid of parts available in catalog supplied by Sachin
- D. Pid of parts available in catalog supplied by all except Sachin

Answer key



7.18.11 SQL: ISRO-DEC2017-11

Consider the schema

Sailors(sid,sname,rating,age) with the following data

sid	sname	rating	age
22	Dustin	7	45
29	Borg	1	33
31	Pathy	8	55
32	Robert	8	25
58	Raghu	10	17
64	Herald	7	35
71	Vishnu	10	16
74	King	9	35
85	Archer	3	26
84	Bob	3	64
96	Flinch	3	17

For the query

```
SELECT S.rating, AVG(S.age) AS avgage FROM Sailors S
Where S.age >= 18
GROUP BY S.rating
HAVING 1<(SELECT COUNT(*) FROM$ Sailors S2 Where S.rating=S2.rating)
```

The number of rows returned is

A. 6

B. 5

C. 4

D. 3

isrodec2017 sql databases

Answer key

7.18.12 SQL: ISRO2015-19



The maximum length of an attribute of type text is

A. 127

B. 255

C. 256

D. It is variable

isro2015 databases sql

Answer key

7.18.13 SQL: ISRO2015-22



Consider the following relational query on the above database:

```
SELECT S.name FROM Suppliers S
Where S.sid NOT IN (SELECT C.sid FROM Catalog C
WHERE C.pid NOT IN (SELECT P.pid FROM Parts P WHERE P.color <>'blue'))
```

- A. Find the names of all suppliers who have supplied non-blue part.
- B. Find the names of all suppliers who have not supplied non-blue part.
- C. Find the names of all suppliers who have supplied only non-blue parts
- D. Find the names of all suppliers who have not supplied only non-blue part.

isro2015 databases sql ambiguous

Answer key

7.18.14 SQL: ISRO2015-23



Consider the following schema:

Emp (Empcode, Name, Sex, Salary, Deptt)

A simple SQL query is executed as follows:

```
SELECT Deptt FROM Emp
GROUP by Dept
Having avg (Salary) > {select avg (Salary) from Emp}
```

The output will be

- A. Average salary of male employee is the average salary of the organization
- B. Average salary of male employee is less than the average salary of the organization
- C. Average salary of male employee is equal to the average salary of the organization
- D. Average salary of male employee is more than the average salary of the organization

isro2015 databases sql

Answer key

7.19

Transaction and Concurrency (12)

7.19.1 Transaction and Concurrency: GATE CSE 2003 | Question: 29, ISRO2009-73



Which of the following scenarios may lead to an irrecoverable error in a database system?

- A. A transaction writes a data item after it is read by an uncommitted transaction
- B. A transaction reads a data item after it is read by an uncommitted transaction
- C. A transaction reads a data item after it is written by a committed transaction
- D. A transaction reads a data item after it is written by an uncommitted transaction

gatecse-2003 databases transaction-and-concurrency easy isro2009

Answer key 

7.19.2 Transaction and Concurrency: GATE CSE 2006 | Question: 20, ISRO2015-17



Consider the following log sequence of two transactions on a bank account, with initial balance 12000, that transfer 2000 to a mortgage payment and then apply a 5% interest.

1. T1 start
2. T1 B old = 12000 new = 10000
3. T1 M old = 0 new = 2000
4. T1 commit
5. T2 start
6. T2 B old = 10000 new = 10500
7. T2 commit

Suppose the database system crashes just before log record 7 is written. When the system is restarted, which one statement is true of the recovery procedure?

- A. We must redo log record 6 to set B to 10500
- B. We must undo log record 6 to set B to 10000 and then redo log records 2 and 3
- C. We need not redo log records 2 and 3 because transaction T1 has committed
- D. We can apply redo and undo operations in arbitrary order because they are idempotent

gatecse-2006 databases transaction-and-concurrency normal isro2015

Answer key 

7.19.3 Transaction and Concurrency: ISRO 2017-5



Which of the following concurrency control protocol ensures both conflict and free from deadlock? ,

- A. Time stamp ordering
- B. 2 Phase locking
- C. Both (a) and (b)
- D. None of the above

isro2017 databases transaction-and-concurrency

Answer key 

7.19.4 Transaction and Concurrency: ISRO CSE 2007 | Question: 60



Which of the following is correct with respect to Two phase commit protocol?

- A. Ensures serializability
- B. Prevents Deadlock
- C. Detects Deadlock
- D. Recover from Deadlock

isro2007 databases transaction-and-concurrency

Answer key 

7.19.5 Transaction and Concurrency: ISRO CSE 2009 | Question: 71



Which of the following contains complete record of all activity that affected the contents of a database during a certain period of time?

- A. Transaction log
- C. Report writer

- B. Query language
- D. Data manipulation language

isro2009 databases transaction-and-concurrency

Answer key 

7.19.6 Transaction and Concurrency: ISRO CSE 2011 | Question: 15



What is the equivalent serial schedule for the following transactions?

Transaction

T1	T2	T3
		R(Y)
		R(Z)
R(X)		
W(X)		
		W(Y)
		W(Z)
	W(Z)	
R(Y)		
W(Y)		

	R(Y)	
	W(Y)	
	R(X)	
	W(X)	

- A. $T_1 - T_2 - T_3$
C. $T_2 - T_1 - T_3$

- B. $T_3 - T_1 - T_2$
D. $T_1 - T_3 - T_2$

isro2011 databases transaction-and-concurrency

Answer key 

7.19.7 Transaction and Concurrency: ISRO CSE 2016 | Question: 58



Trigger is

- A. Statement that enables to start any DBMS
B. Statement that is executed by the user when debugging an application program
C. The condition that the system tests for the validity of the database user
D. Statement that is executed automatically by the system as a side effect of a modification of the database

isro2016 databases transaction-and-concurrency

Answer key 

7.19.8 Transaction and Concurrency: ISRO CSE 2017 | Question: 6



ACID properties of a transactions are

- A. Atomicity, consistency, isolation, database
B. Atomicity, consistency, isolation, durability
C. Atomicity, consistency, integrity, durability
D. Atomicity, consistency, integrity, database

isro2017 databases transaction-and-concurrency

Answer key 

7.19.9 Transaction and Concurrency: ISRO CSE 2018 | Question: 55



Let us assume that transaction T_1 has arrived before transaction T_2 . Consider the schedule $s = r_1(A); r_2(B); w_2(A); w_1(B)$

Which of the following is true?

- a. Allowed under basic timestamp protocol.
- b. Not allowed under basic timestamp protocols because T_1 is rolled back.
- c. Not allowed under basic timestamp protocols because T_2 is rolled back.
- d. None of these

isro2018 databases transaction-and-concurrency

Answer key

7.19.10 Transaction and Concurrency: ISRO CSE 2020 | Question: 22



Raymonds tree based algorithm ensures

- A. no starvation, but deadlock may occur in rare cases
- B. no deadlock, but starvation may occur
- C. neither deadlock nor starvation can occur
- D. deadlock may occur in cases where the process is already starved

isro-2020 databases transaction-and-concurrency normal

Answer key

7.19.11 Transaction and Concurrency: ISRO CSE 2023 | Question: 49



Amongst the ACID properties of a transaction, the 'Durability' property requires that the changes made to the database by a successful transaction persist

- A. Except in case of an operating system crash
- B. Except in case of a disk crash
- C. Except in case of a power failure
- D. Always, even if there is a failure of any kind

isro-cse-2023 databases transaction-and-concurrency

Answer key

7.19.12 Transaction and Concurrency: ISRO-2013-23



Which of the following is the highest isolation level in transaction management?

- A. Serializable
- B. Repeated Read
- C. Committed Read
- D. Uncommitted Read

isro2013 transaction-and-concurrency

Answer key

Answer Keys

7.0.1	B	7.0.2	C	7.1.1	A	7.1.2	A	7.1.3	B
7.1.4	C	7.1.5	A	7.1.6	B	7.2.1	D	7.3.1	B
7.4.1	D	7.4.2	C	7.4.3	B	7.4.4	B	7.4.5	A
7.4.6	B	7.4.7	C	7.4.8	A	7.4.9	D	7.4.10	C
7.4.11	C	7.4.12	Q-Q	7.4.13	B	7.5.1	A	7.6.1	A

7.7.1	D
7.8.1	A
7.8.6	A
7.10.4	A
7.14.1	B
7.16.2	C
7.18.1	D
7.18.6	C
7.18.11	D
7.19.2	B
7.19.7	D
7.19.12	A

7.7.2	B
7.8.2	C
7.9.1	B
7.11.1	B
7.14.2	C
7.16.3	A
7.18.2	B
7.18.7	A;C
7.18.12	D
7.19.3	A
7.19.8	B

7.7.3	B
7.8.3	D
7.10.1	A
7.12.1	B
7.15.1	B
7.16.4	A
7.18.3	A
7.18.8	A
7.18.13	X
7.19.4	A
7.19.9	B

7.7.4	A
7.8.4	D
7.10.2	C
7.13.1	C
7.15.2	B
7.17.1	B
7.18.4	B
7.18.9	A
7.18.14	D
7.19.5	A
7.19.10	B

7.7.5	C
7.8.5	B
7.10.3	A
7.13.2	C
7.16.1	B
7.17.2	B
7.18.5	A
7.18.10	B
7.19.1	D
7.19.6	B
7.19.11	D